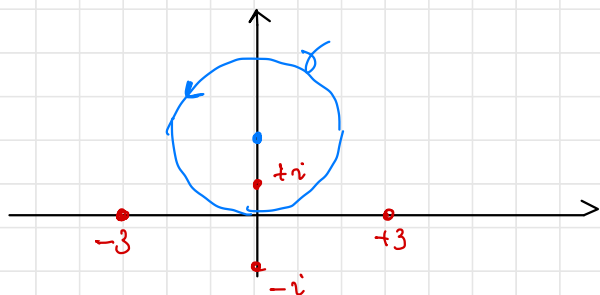


Esempio

$$\int_{C_2(2i)} \overbrace{\frac{f(z)}{1}}^{f(z)} dz = 2\pi i \sum_{z_0 \in S} \text{Res}(f, z_0) \text{Ind}(\gamma, z_0)$$



$$\Omega = \emptyset$$

$$S = \{\pm i, \pm 3\}$$

$$\text{acc}(S) \cap \Omega = \emptyset$$

$$S \cap C_2(2i) = \emptyset$$

$$\text{Ind}(\gamma, \pm 3) = \text{Ind}(\gamma, -i) = 0.$$

$$\text{Ind}(\gamma, i) = 1$$

$$\text{Res}(f, i) = \lim_{z \rightarrow i} (z - i) f(z)$$

↑
i polo semplice (ordine 1)

$$= \lim_{z \rightarrow i} \frac{1}{(z+i)(z^2-9)} = \dots$$

Esempi di applicazioni in campo reale.

Tipo 1

$$\int_0^{2\pi} f(\cos t, \sin t) dt \stackrel{(\circ)}{=} \int_0^{2\pi} \underbrace{g\left(\underbrace{e^{it}}_{r(t)}\right)}_{r'(t)} \underbrace{ie^{it}}_{r'(t)} dt =$$

$$\stackrel{\uparrow}{=} \int_{C_1(0)} g(z) dz \stackrel{\uparrow}{=} 2\pi i \sum_{\substack{z_0 \in S(g) \\ |z_0| < 1}} \text{Res}(g, z_0)$$

TEO RESIDUI

g soddisfa ip. - treo residui su $\Omega \supseteq C_1(0)$, $\gamma = C_1(0)$

Esempio

$$\int_0^{2\pi} \frac{1}{2 + \sin t} dt \stackrel{\uparrow}{=} \int_0^{2\pi} \frac{1}{2 + \frac{e^{it} - e^{-it}}{2i}} dt$$

$$\sin t = \frac{e^{it} - e^{-it}}{2i}$$

$$= \int_0^{2\pi} \frac{2i}{4i + e^{it} + e^{-it}} \cdot \frac{ie^{it}}{ie^{it}} dt = g(e^{it})$$

$$e^{it} = z \Rightarrow g(z) = \frac{2i}{4i + z + z^{-1}} \cdot \frac{1}{z} = \frac{2}{z^2 + 4iz - 1}$$

$\uparrow$

$$= 2\pi i \sum_{\substack{z_0 \in S(g) \\ |z_0| < 1}} \text{Res}(g, z_0)$$

Valor principale

(V.P.)

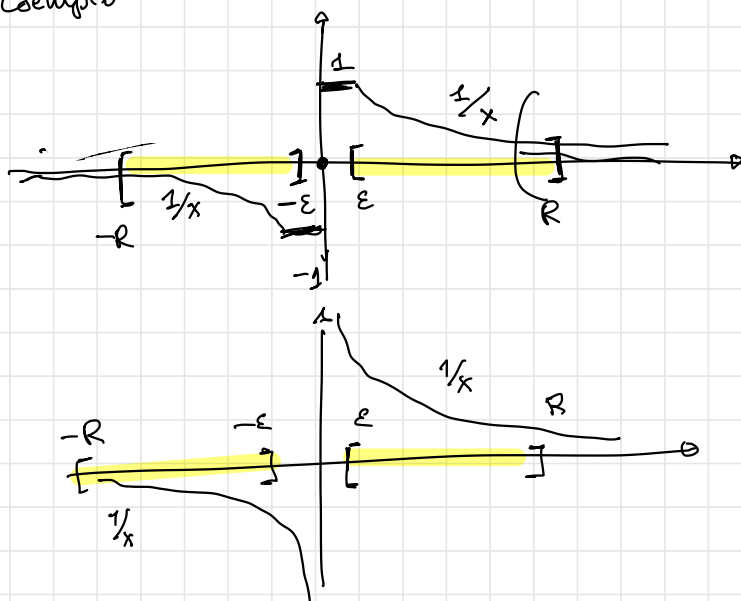
$$\int_{\mathbb{R}} f(x) dx := \lim_{\substack{R \rightarrow +\infty \\ \varepsilon \rightarrow 0}} \int_{-R}^{-\varepsilon} f(x) dx + \int_{\varepsilon}^R f(x) dx$$

↑
limitata
vicino a 0

$R \rightarrow +\infty$
 $\varepsilon \rightarrow 0$

$$\int_{-R}^{-\varepsilon} f(x) dx + \int_{\varepsilon}^R f(x) dx$$

Esempio

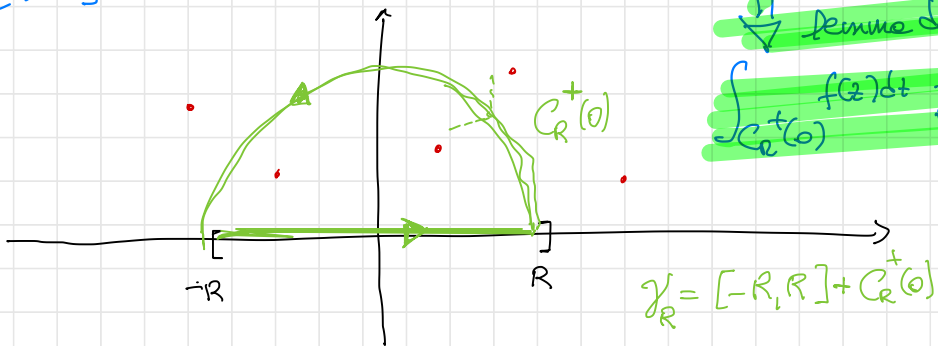


Tipo 2

$$I = \text{V.P.} \int_{\mathbb{R}} f(x) dx = 2\pi i \sum_{\substack{z_0 \in S(f) \\ \text{Im } z_0 > 0}} \text{Res}(f, z_0)$$

IPOTESI $f = f(z)$ abbia un n° finito di singolarità $\{ \text{Im } z > 0 \}$
e NON abbia singolarità sull'asse reale

(*) $\alpha > 1$
 $C > 0$ tale che per $|z| \gg 1$, $|f(z)| \leq \frac{C}{|z|^\alpha}$



$$I = \lim_{R \rightarrow +\infty} \left\{ \underbrace{\left[\int_{-R}^R f + \int_{C_R^+(0)} f \right]}_{\text{"}} - \int_{C_R^+(0)} f \right\}$$

$$\int_{\gamma_R} f = 2\pi i \sum_{\substack{z_0 \in S(f) \\ \text{Im } z_0 > 0 \\ |z_0| < R}} \text{Res}(f, z_0).$$

$$\downarrow R \rightarrow +\infty$$

$$2\pi i \sum_{\substack{z_0 \in S(f) \\ \text{Im } z_0 > 0}} \text{Res}(f, z_0)$$

Esempio

$$I = \int_{\mathbb{R}} \frac{1}{1+x^2} dx = 2\pi i \operatorname{Res}(f, i)$$

$$\text{"} \left. \arctan x \right|_{-\infty}^{+\infty} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

$$f(z) = \frac{1}{1+z^2}$$

$$S(f) = \pm i$$

Tipo 3

$$I = \text{V.P.} \int_{\mathbb{R}} f(x) e^{i\omega x} dx = 2\pi i \sum_{\substack{z_0 \in S(f) \\ \text{Im } z_0 > 0}} \text{Res}(f(z) e^{i\omega z}, z_0)$$

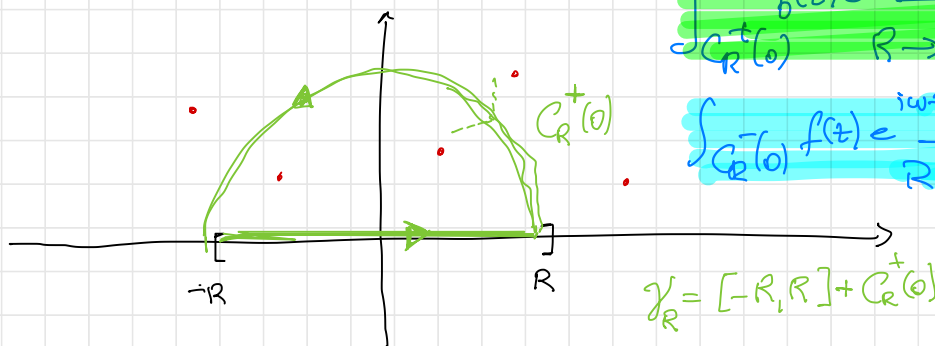
$\in \mathbb{R}^+$ $\in \mathbb{R}^-$

IPOTESI $f = f(z)$ abbia un n° finito di singolarità $\{ \text{Im } z > 0 \}$
 e NON abbia singolarità sull'asse reale

(**) $\lim_{R \rightarrow +\infty} \sup_{z \in C_R^+(0)} |f(z)| = 0 \implies$ Lemma di Jordan

$$\int_{C_R^+(0)} f(z) e^{i\omega z} dz \xrightarrow{R \rightarrow +\infty} 0$$

$$\int_{C_R^-(0)} f(z) e^{i\omega z} dz \xrightarrow{R \rightarrow +\infty} 0$$



$$\gamma_R = [-R, R] + C_R^+(0)$$

$$I = \lim_{R \rightarrow +\infty} \left\{ \underbrace{\left[\int_{-R}^R f + \int_{C_R^+(0)} f \right]}_{\int_{\gamma_R} f} - \int_{C_R^+(0)} f \right\}$$

$$\int_{\gamma_R} f = 2\pi i \sum_{\substack{z_0 \in S(f) \\ \text{Im } z_0 > 0 \\ |z_0| < R}} \text{Res}(f, z_0).$$

$$\downarrow R \rightarrow +\infty$$

$$2\pi i \sum_{\substack{z_0 \in S(f) \\ \text{Im } z_0 > 0}} \text{Res}(f, z_0)$$

Example

$$I = \text{v.p.} \int \frac{e^{ix}}{1+x^2} dx \quad \omega=1 \\ = 2\pi i \operatorname{Res} \left(f(z) e^{i\omega z}, i \right)$$

$$S(f) = \{i\} \quad \text{no singularities on } \mathbb{R}$$

$$\sup_{C_R^+(0)} \left| \frac{1}{1+z^2} \right| \xrightarrow{R \rightarrow +\infty} 0 \implies \text{poss. applicare Jordan}$$

$$\sup_{C_R^+(0)} \frac{1}{|1+z^2|} = \frac{1}{\inf_{C_R^+(0)} |1+z^2|}$$

$$\inf_{C_R^+(0)} |1+z^2| \xrightarrow{R \rightarrow +\infty} +\infty$$

$$|1+z^2| \geq |z^2| - 1 = R^2 - 1$$

$$|z^2| = |z^2 + 1 - 1| \leq |z^2 + 1| + 1$$

$$|1+z^2| \geq R^2 - 1 \quad \forall z \in C_R^+(0)$$

$$\Rightarrow \inf_{z \in C_R^+(0)} |1+z^2| \geq R^2 - 1 \longrightarrow +\infty$$

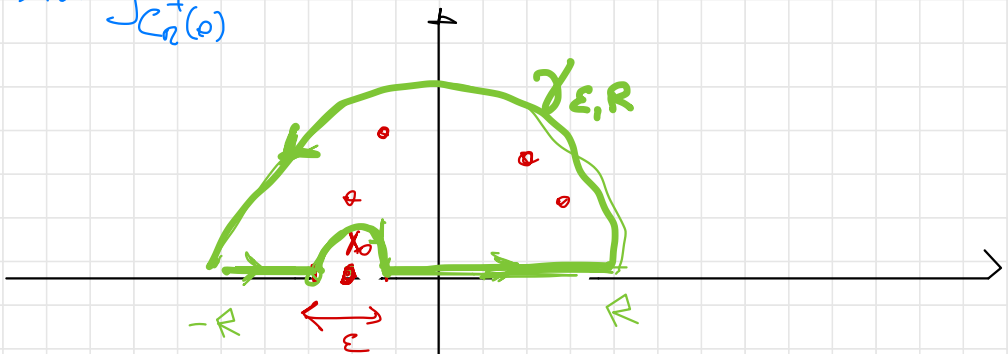
Tipo 4

$$I = \text{V.P.} \int_{\mathbb{R}} f(x) dx = 2\pi i \sum_{\substack{z_0 \in S(f) \\ \text{Im} z_0 > 0}} \text{Res}(f, z_0) + \pi i \sum_{\substack{z_0 \in S(f) \\ z_0 \in \mathbb{R}}} \text{Res}(f, z_0)$$

IPOTESI

$f = f(z)$ abbia un n° finito di singolarità su $\{ \text{Im} z > 0 \}$
e un n° finito di singolarità su $\mathbb{R}$. (POLISEMPLICE)

$$\lim_{R \rightarrow +\infty} \int_{C_R^+(0)} f(z) dz = 0$$



$$I = \lim_{\substack{R \rightarrow +\infty \\ \epsilon \rightarrow 0}} \int_{-R}^{x_0 - \epsilon} f + \int_{x_0 + \epsilon}^R f$$

$$= \lim_{\substack{R \rightarrow +\infty \\ \epsilon \rightarrow 0}} \int_{\gamma_{\epsilon, R}} f - \lim_{R \rightarrow +\infty} \int_{C_R^+(0)} f + \lim_{R \rightarrow +\infty} \int_{C_\epsilon^+(x_0)} f$$

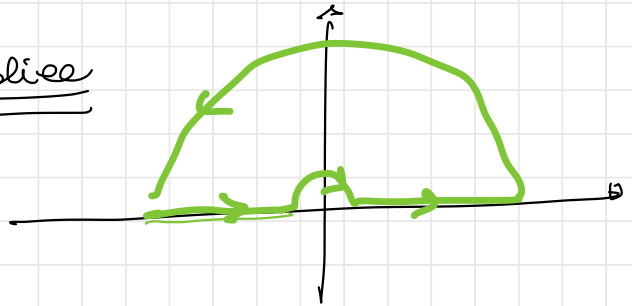
Lemme x_0 polo semplice $\Rightarrow \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon^+(x_0)} f = \pi i \text{Res}(f, x_0)$

Esempio

$$I = \text{v.p.} \int_{\mathbb{R}} \frac{1 - e^{2ix}}{x^2} dx = \pi i \operatorname{Res}(f, 0)$$

$$f(z) = \frac{1 - e^{2iz}}{z^2} \quad \text{olomorfe su } \mathbb{C} \setminus \{0\}$$

$z^0 = 0$ ipolo semplice



$$\lim_{R \rightarrow +\infty} \int_{C_R^+(0)} f(z) dz = 0 \quad (?)$$

$$f(z) = \underbrace{\frac{1}{z^2}}_{f_1(z)} - \underbrace{\frac{e^{2iz}}{z^2}}_{f_2(z) = g(z)e^{i\omega z}} \quad \omega = 2 \quad g(z) = \frac{1}{z^2}$$

$$\lim_{R \rightarrow +\infty} \int_{C_R^+(0)} \frac{1}{z^2} dz = 0 \quad \text{Lemma decadimento} \quad (\alpha = 2)$$

$$\lim_{R \rightarrow +\infty} \int_{C_R^+(0)} \frac{1}{z^2} e^{2iz} dz = 0 \quad \text{Lemma Jordan (su } C_R^+(0))$$

$$\frac{1}{R^2} = \sup_{C_R^+(0)} \left| \frac{1}{z^2} \right| \rightarrow 0$$